

CARLSBAD CAVERN CITY, NM, USA

WMO: 722687

Lat: 32.334N

Lon: 104.258W

Elev: 3232

StdP: 13.06

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 93033

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	20.6	24.2	-1.5	5.7	47.6	2.2	6.9	49.2	33.7	55.8	29.1	56.6	5.6	310	0.518

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.5	103.2	65.0	100.4	65.3	98.3	65.3	71.6	86.1	70.7	85.8	69.7	85.5	11.1	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.6	118.4	74.9	67.2	112.8	74.4	65.9	107.6	74.4	37.5	85.5	36.5	85.7	35.7	85.2	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.2	24.3	20.3	13.2	108.1	5.0	2.2	9.6	109.7	6.7	111.0	3.8	112.3	0.2	113.9
			11.4	74.0	4.6	1.9	8.1	75.4	5.5	76.5	2.9	77.6	-0.4	79.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.3	44.8	49.3	56.4	64.5	73.5	81.8	83.0	81.6	74.7	64.1	52.5	44.3		
	DBStd	15.75	8.52	9.18	8.54	7.79	6.96	4.64	4.12	4.13	5.73	7.44	8.08	8.48		
	HDD50	647	203	111	39	6	0	0	0	0	0	6	66	216		
	HDD65	2595	625	441	282	100	17	0	0	0	6	104	378	642		
	CDD50	5869	43	92	238	442	730	955	1022	981	741	444	142	39		
	CDD65	2343	0	2	16	86	282	505	557	516	297	78	4	0		
	CDH74	30893	18	115	537	1594	3909	6901	7137	6244	3178	1091	155	14		
	CDH80	16793	1	19	148	653	2115	4207	4161	3505	1557	404	23	0		
(14) Wind	WSAvg	9.0	8.8	9.7	10.4	11.1	10.2	9.6	8.1	7.4	7.7	8.2	8.3	8.5		
(15) Precipitation	PrecAvg	13.0	0.4	0.5	0.4	0.6	1.1	1.5	1.7	2.0	2.4	1.2	0.7	0.6		
	PrecMax	24.2	1.7	2.2	2.2	4.2	4.2	5.7	7.5	7.7	12.3	5.8	4.6	3.8		
	PrecMin	4.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	5.0	0.4	0.6	0.5	0.9	1.1	1.4	1.7	1.9	2.6	1.4	1.1	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.6	83.0	88.2	94.4	102.1	108.0	105.7	103.8	99.7	92.5	83.1	76.1		
		MCWB	50.6	53.0	54.7	57.7	63.0	65.7	66.4	67.2	65.6	61.2	55.3	50.6		
	2%	DB	72.2	77.5	84.1	90.2	98.7	104.3	101.8	100.0	95.6	88.5	78.6	71.6		
		MCWB	48.7	50.8	53.4	56.0	61.5	63.9	66.8	66.9	64.6	60.8	54.4	49.2		
	5%	DB	67.7	73.1	80.5	87.2	95.2	101.4	99.3	97.5	92.8	84.6	74.4	66.4		
		MCWB	47.3	49.1	52.2	55.1	60.0	63.9	66.9	67.0	64.9	60.0	52.2	47.2		
	10%	DB	62.9	68.3	75.8	83.9	91.4	98.1	96.7	95.1	89.3	80.6	70.0	61.6		
		MCWB	44.9	47.4	50.6	54.5	59.1	64.2	66.7	67.4	64.9	59.0	51.1	45.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.4	55.6	60.0	63.7	68.6	71.9	73.5	73.1	72.2	70.2	60.4	53.7		
		MCDB	68.5	73.1	71.4	77.1	87.1	90.9	85.3	86.3	80.9	78.3	71.2	68.5		
	2%	WB	50.3	53.0	56.6	60.9	66.6	70.3	71.4	71.4	70.2	67.0	57.0	50.7		
		MCDB	67.4	72.5	74.5	76.4	84.4	88.9	86.8	85.6	82.3	77.3	71.4	66.7		
	5%	WB	48.0	50.6	54.3	58.9	65.1	69.1	70.5	70.5	68.9	64.5	54.8	48.4		
		MCDB	65.6	69.6	73.7	76.6	83.2	87.5	86.3	85.5	82.0	75.0	69.9	64.3		
	10%	WB	45.6	48.4	52.2	56.5	63.5	68.1	69.5	69.5	67.8	62.1	52.4	46.0		
		MCDB	61.4	66.7	72.0	77.7	81.3	86.8	85.8	85.4	80.8	74.1	67.2	60.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	28.2	29.0	30.2	30.9	29.7	29.5	25.5	25.0	25.2	27.6	28.7	26.6		
		MCDBR	36.6	37.8	38.4	36.9	35.6	35.7	29.6	28.8	31.1	34.5	36.6	35.4		
		MCWBR	19.3	18.4	16.8	15.7	14.5	11.5	8.3	8.2	9.9	14.1	17.7	18.4		
		MCDBR	33.8	34.5	33.7	31.8	30.2	28.7	24.5	24.1	23.7	25.3	31.2	31.9		
	5% WB	MCWBR	18.7	17.3	15.7	14.7	13.2	9.7	6.9	7.1	8.1	11.2	16.0	17.1		
(40) Clear-Sky Solar Irradiance	taub		0.253	0.262	0.280	0.317	0.331	0.357	0.383	0.364	0.341	0.295	0.273	0.259		
	taud		2.616	2.591	2.530	2.401	2.428	2.404	2.377	2.423	2.441	2.578	2.566	2.591		
	Ebn at Noon		315	320	319	308	303	293	285	290	293	304	303	306		
	Edh at Noon		24	27	31	37	37	38	39	36	34	28	25	23		
(44) All-Sky Solar Radiation	RadAvg		1128	1422	1829	2238	2406	2455	2292	2108	1821	1499	1182	1001		
	RadStd		98	121	112	131	135	107	109	91	118	134	106	76		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (12 neighbors)	N/A	-1.31	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page